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Diff Studio: Ecosystem for Interactive Modeling by Ordinary Differential Equations

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Software

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Summary

Ordinary differential equations (ODEs) are crucial in modeling complex systems and phenomena. Their applications range from pharmacology and drug manufacturing to financial modeling and environmental studies.

Diff Studio is a high-performance TypeScript application for solving initial value problems (IVPs) for ODEs directly within web browsers. It consists of two components. The **Diff Grok library** implements numerical methods and formula parsing tools. The **Diff Studio application** integrates Diff Grok tools with **Datagrok**, a scientific computing platform free for personal and academic use.

Diff Studio provides an ecosystem for rapid development of ODE-based applications with reproducible and accessible models.

Statement of need

Scientific modeling of complex processes and phenomena uses ODEs. They are widely applied in diverse fields, including physical processes (Chicone, 2006), biochemical kinetics (Ingalls, 2013), drug delivery systems (Mircioiu et al., 2019), cloud computing (Ghomi et al., 2019), and population dynamics (Hastings, 2013).

Analytic methods providing exact solutions can be applied only to a limited class of ODEs. The use of analytic solutions often proves impractical due to their complexity (Hairer et al., 2008). Numerical methods computing approximate solutions are often preferred.

Many methods for solving ODEs have been recently developed (Hairer et al., 2008; Hairer & Wanner, 2002). These methods have been implemented in various software tools, including libraries and packages for programming languages and scientific computing environments. Notable examples include SUNDIALS (Gardner et al., 2022; Hindmarsh et al., 2005), Julia Differential Equations package (Rackauckas & Nie, 2017), SciPy (Virtanen et al., 2020), Maple (Maplesoft, 2025), Mathematica (Wolfram Research Inc., 2024), Matlab (The MathWorks Inc., 2022), and deSolve (Soetaert et al., 2010).

Most tools require expertise, shifting focus from research to development. The goal of this project is to develop an ecosystem providing a combination of a "no-code" approach with comprehensive capabilities for in-browser modeling and analysis.

The solution: Diff Studio

Diff Grok library

This library provides numerical methods and automatic generation of JavaScript code from a declarative problem specification. It includes:

- **Solving tools:** numerical methods for solving IVPs;
- **Scripting tools:** methods for automatic generation of JavaScript code that solves problems specified in the declarative form.

Solving tools implement:

- `mrt` - Modified Rosenbrock triple (MRT) (Shampine & Reichelt, 1997)
- `ros3prw` - the ROS3PRw method (Jax et al., 2021)
- `ros34prw` - the ROS34PRw method (Rang, 2015)

To solve

$$\begin{aligned} dy/dt &= f(t, y), \\ y(t_0) &= y_0 \end{aligned} \tag{1}$$

on the interval $[t_0, t_1]$, define an ODEs object. This object specifies the independent variable (t), its range ($[t_0, t_1]$), solution grid step size (h), initial conditions (y_0), right-hand side of the ODEs, tolerance, and names of dependent variables. Next, apply a selected method (`mrt`, `ros3prw` or `ros34prw`) to this object. The output consists of a list of `float64` arrays containing the values of the independent variable and the corresponding approximate solutions.

For example, consider

$$\begin{aligned} dx/dt &= x + y - t, \\ dy/dt &= xy + t \\ x(0) = 1, y(0) &= -1 \\ t \in [0, 2], h &= 0.001 \end{aligned} \tag{2}$$

In this case, the ODEs object is defined as follows:

```
const task: ODEs = {
    name: 'Example',
    arg: {
        name: 't',
        start: 0,
        finish: 2,
        step: 0.001,
    },
    initial: [1, -1],
    func: (t: number, y: Float64Array, output: Float64Array) => {
        output[0] = y[0] + y[1] - t;
        output[1] = y[0] * y[1] + t;
    },
    tolerance: 1e-7,
    solutionColNames: ['x', 'y'],
};
```

The following code solves the given problem:

```
const solution = mrt(task);
```

The solution contains three items:

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- `solution[0]` - values of `t`, i.e., the range `0..2` with the step `0.001`;
- `solution[1]` - values of `x(t)` at the points of this range;
- `solution[2]` - values of `y(t)` at the same points.

Diff Grok delivers outstanding computational performance (Datagrok Inc, 2025b), benchmarked on **Robertson** (Robertson, 1966), **HIRES** (Schäfer, 1975), **VDPOL** (Pol (van der), 1926), **OREGO** (Hairer & Wanner, 2002), **E5** (Hairer & Wanner, 2002), and **Pollution** (Verwer, 1994). It allows users to obtain the modeling results in near-real time.

Scripting tools enable specification of IVPs declaratively using an intuitive syntax (see Figure 1).

```
#name: Example

#equations:
  dx/dt = x + y - t
  dy/dt = x * y + t

#argument: t
  initial = 0
  final = 2
  step = 0.001

#inits:
  x = 1
  y = -1

#tolerance: 1e-7
```

Figure 1: Diff Studio model corresponding to the Equation 2.

The method `getIVP()` parses a model and produces the IVP object specifying the problem. The method `getJScode()` generates JavaScript code, involving an appropriate ODEs object, that can be applied for solving equations.

Diff Studio package

The Diff Studio package integrates Diff Grok with Datagrok (Datagrok Inc, 2025a). It implements an application with a model editor (Figure 2).

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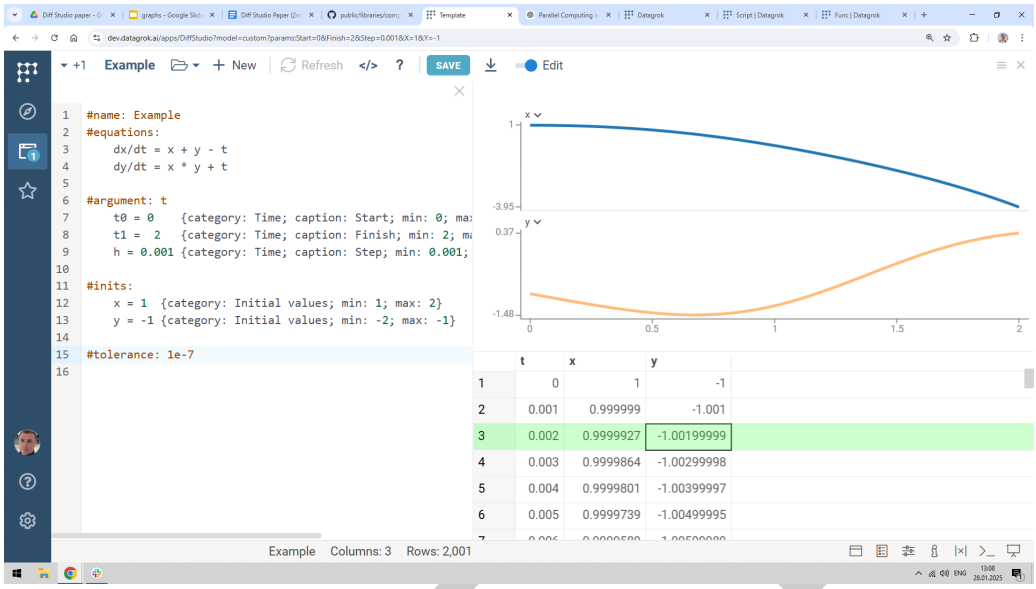

Figure 2: The Diff Studio application: the equation editor, numerical solution of the problem Equation 2, and its visualization.

Diff Studio automatically generates the user interface (Figure 3). Each model input can be annotated using self-explanatory options (Figure 4).

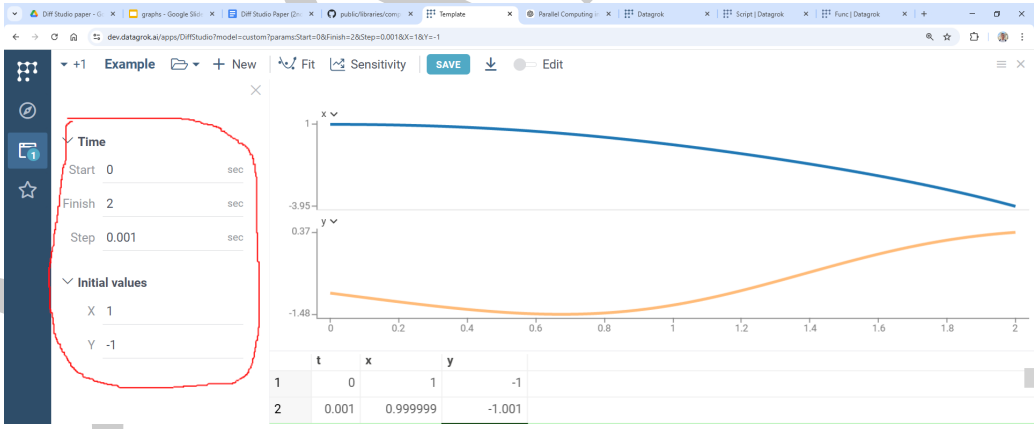

Figure 3: The Diff Studio application, Autogenerated UI: Diffstudio creates input entries (highlighted) for all variables listed in the equation editor. Each time model inputs are changed, a solution is computed and displayed.

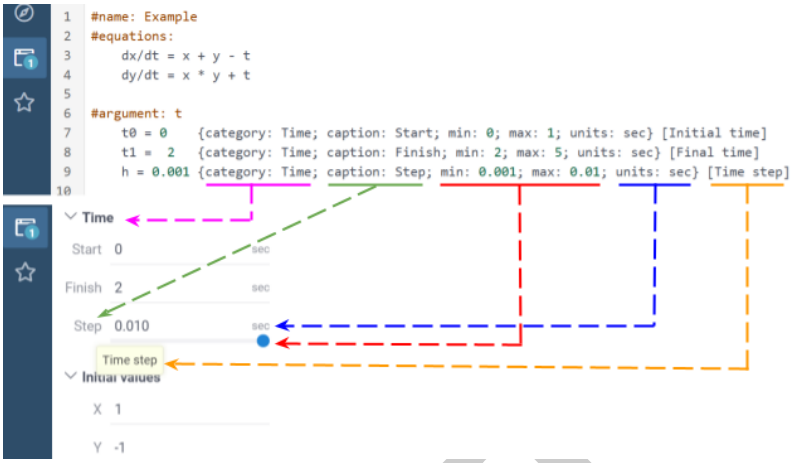

Figure 4: The correspondence of input annotation from Figure 2 and UI elements from Figure 3

Datagrok provides in-browser computations and visualizations. Other features include:

- Sensitivity analysis and parameter optimization;
- Storing and sharing computations.

Thus, Diff Studio serves as a comprehensive modeling environment.

Availability

Diff Grok is available on GitHub.

The Diff Studio package is accessible on GitHub, while its documentation can be found at Datagrok Help pages.

Run Diff Studio online here, or complete an interactive tutorial.

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Conflicts of interest

The authors declare no conflict of interest.

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